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Charge behaviour and power consumption in ball mills: sensitivity to mill operating conditions, liner geometry and charge composition

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Abstract

Discrete element method (DEM) modelling has been used to systematically study the effects of changes in mill operating parameters and particle properties on the charge shape and power draw of a 5-m ball mill. Specifically, changes in charge fill level, lifter shape (either by design or wear) and lifter pattern are analysed. The effects of changes to the properties of the charge (ball fraction, ball and rock shape, type of ball and rock size distributions and the lower cutoff of the rock size distribution) can all be interpreted in terms of their effects on the shear strength of the charge. Some changes increase the shear strength leading to higher dynamic angles of repose of the charge, higher shoulder positions and higher power consumption for sub-critical speeds. For super-critical speeds, they lead to lower power consumption, due to lower particle mobility as the particles lock together better. Changes to the charge that weaken the interlocking of particles have the opposite effect on the charge shape and power consumption. The combination of these effects means that the speed for which peak power consumption occurs is predominantly determined by the shear strength of the charge material and the fill level. This demonstrates the sensitivity of mill behaviour to the charge characteristics and the critical importance of various assumptions used in DEM modelling. © 2001 Elsevier Science B.V. All rights reserved.

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1. Introduction

Ball mills are used in mineral processing for particle size reduction. These consist of a rotating cylindrical shell of up to 7 m in diameter with axial lengths greater than this. Lifter bars and sacrificial liner plates are bolted to the inside of the mill shell. Rock generally arrives from a crusher or perhaps a semi-autogenous (SAG) mill (depending on the design of the milling circuit and the mineral being processed) and enters the feed end of the mill. Grinding media consisting of steel balls, with typical diameters from 25 to 75 mm, is already present in the mill along with any rock that has not yet overflowed through the discharge end. Typical power consumption for a 5-m diameter by 7-m-long ball mill is between 2.5 and 3.5 MW. The actual proportion of this energy usefully used in size reduction is thought to be very low, perhaps in the range of 1–5%. Significant financial and environmental benefits can be obtained by improving this efficiency even slightly. Substantial benefits are also potentially available through higher downstream recoveries of the liberated mineral if the product particle size distributions (PSD) can be tailored for subsequent flotation or other separation processes. Large saving would also be possible if the lifespan of the mill liners could be extended by reducing peak wear rates.

Microscale simulation of ball and other types of mills by discrete element methods (DEM) offers the opportunity of better understanding the internal mill dynamics and for developing improvements to mill design and operation that can lead to large increases in mill efficiency and throughput. DEM modelling has been used for a very wide range of applications in geophysics (such as landslides by Cleary and Campbell, 1993; Campbell et al., 1995 and ice flows by Hopkins et al., 1991) and industry (ball mills by Mishra and Rajamani, 1992, 1994; Cleary, 1998a, SAG mills by Rajamani and Misha, 1996 and Cleary, 2001, vibrational segregation by Haff and Werner, 1986; Ristow, 1994 and Cleary, 1998c, rotating drums by Ristow, 1994; Pöschel and Buchholtz, 1995 and others, sampling by Robinson and Cleary, 1999, draglines by Cleary, 1998b, 2000, hopper flows by Holst et al., 1999; Ristow, 1994 and many more). Of particular relevance is experimental validation of DEM predictions for a centrifugal mill by Cleary and Hoyer (2000), which showed excellent agreement both for the charge motion and the power draw.

This paper is the second installment of an ongoing numerical study of ball mills. The first installment (Cleary, 1998a) described the basic modelling of a 5-m ball mill. Predictions of charge profile, charge segregation, power draw, torque, liner wear and collisional force distributions were made for a range of mill conditions. The sensitivity of the power draw to some particle properties and some operating parameters was analysed in a preliminary way. The power draw was found to be relatively insensitive to material properties such as the coefficients of restitution and friction, with variations generally around 5% or less for normal operating conditions.

An important observation from this work was that the behaviour of a charge consisting of balls and rocks was reasonably different to that of a charge consisting purely of balls (as frequently used in earlier DEM simulation, Mishra and Rajamani, 1992, 1994). The toe position was found to be higher and the surface profile was more

clearly defined with a more bi-linear shape when the rock fraction was included. The power draw was also found to be noticeably lower.

A simple wear model also allowed predictions of liner wear to be made leading to plausible estimates of lifter life spans. The spatial distributions of wear were also consistent with anecdotal information from mill operations. Finally, two mechanisms were also identified for producing size segregation in mills. One operates at low speed and concentrates large particles around the outside of the charge. The other operates at high speeds and causes large particles to accumulate in the center of the charge. At intermediate speeds, these mechanisms cancel.

In this paper, we analyse the effect of changes in particle size and shape distribution (for both balls and rocks), lifter pattern and profile, and fill level on power draw and charge behaviour for rotation rates between 50% and 130% of the critical speed at which particles begin to centrifuge. All simulation parameters are held constant so that the effect of each of the above influences can be independently evaluated. This study is intended to illustrate the functional dependences of mill power on various mill conditions and charge properties. These dependences are presently still somewhat qualitative, since detailed data for comparison is difficult to obtain and the current DEM model is still somewhat simplified. Detailed validation and the use of large scale 3D DEM should soon allow these to be understood quantitatively as well. The dependences analysed in this paper are based on the relative changes of predicted power with changes in mill design and operational parameters and not on the absolute accuracy of the power predictions. We also evaluate the importance of various assumptions (such as whether particle shape can be neglected) that are made to render feasible the simulation of this complex industrial system. Understanding the effects of different assumptions is critical to developing quantitatively accurate DEM mill modelling.

2. The discrete element method

The essential feature of discrete element simulation is that collisional interactions of particles with each other and with their environment are detected and modelled using a contact force law. Equations of motion are then solved for the particle motions and for the motion of any boundary objects with which the particles interact. In rotating mills, the axial migration of particles is slow compared to the circulating motion in the plane orthogonal to the mill axis. This allows the mill to be plausibly modelled in two dimensions. It is important to be able to model the complex boundary geometry of the liner, including any lifters. When operating in two dimensions our code uses boundary objects that are constructed from segments, which can be line segments, circular segments or discs. Nearly arbitrary shaped two-dimensional mill cross-sections can be constructed.

In this paper, the particles are modelled either as disks or super-quadratics with varying size distributions. Super-quadratics are defined by the equation $x^n + (y/A)^n = s^n$, where the power n determines the blockiness of the resulting particle and A determines the aspect ratio. Aspect ratios of up to 12:1 and blockiness factors of up to 20 can be used.

The general DEM methodology and its variants are well established and are described in review articles by Campbell (1990), Barker (1994) and Walton (1994). Here we use a reasonably standard implementation that is described in more detail in Cleary (1998a,c). Briefly, the particles are allowed to overlap and the amount of overlap Δx , and normal v_n and tangential v_t relative velocities determine the collisional forces via a contact force law. There is a range of possible contact force models available in the literature that approximate the collision dynamics to various extents. We use a linear spring-dashpot model. For more complex models, see Walton (1994) and Schäfer et al. (1996). The normal force:

$$F_n = -k_n \Delta x + C_n v_n, \quad (1)$$

consists of a spring to provide the repulsive force and a dashpot to dissipate a proportion of the relative kinetic energy. The maximum overlap between particles is determined by the stiffness k_n of the spring in the normal direction. Typically, average overlaps of 0.1–1.0% are desirable, so a normal spring constant of 8×10^6 N/m is used in all these simulations. The normal damping coefficient C_n is chosen to give a required coefficient of restitution ε (defined as the ratio of the post-collisional to pre-collisional normal component of the relative velocity):

$$C_n = 2\gamma\sqrt{m_{ij}k_n}, \quad \text{where} \quad \gamma = -\frac{\ln(\varepsilon)}{\sqrt{\pi^2 + \ln^2(\varepsilon)}} \quad \text{and} \quad m_{ij} = \frac{m_i m_j}{m_i + m_j},$$

is the reduced mass of particles i and j with masses m_i and m_j , respectively. This arises from the analytic solution of the normal collision Eq. (1) for two such particles. Each particle can have a different coefficient of restitution (and corresponding damping constant), but we generally use the same value for all the particles because little experimental data on particle to particle variations in ε is available.

The tangential force is given by:

$$F_t = \min\left\{\mu F_n, k_t b \int v_t dt + C_t v_t\right\}, \quad (2)$$

where the integral of the tangential velocity v_t over the collision behaves as an incremental spring that stores energy from the relative tangential motions and represents the elastic tangential deformation of the contacting surfaces. The dashpot dissipates energy from the tangential motion and models the tangential plastic deformation of the contact. The total tangential force (given by the sum of the elastic and plastic components) is limited by the Coulomb frictional limit at which point the surface contact shears and the particles begin to slide over each other. We choose the tangential spring stiffness to be $k_t = \alpha k_n$ and $C_t = \sqrt{\alpha} C_n$. For these contact force laws (Eqs. (1) and (2)) the frictional behaviour in two dimensions is insensitive to the choice of α for values in the range $0.1 \leq \alpha \leq 10$ for both oblique collisions and for sliding along inclined slopes. The value used throughout this study is $\alpha = 10$.

The contact force model requires both the coefficient of elasticity ε and friction μ for the particles to be supplied. Such quantities are in practice very difficult to obtain experimentally and vary from particle to particle and during the flow as the particles are ground. Typical values are therefore chosen, $\varepsilon = 0.3$ and $\mu = 0.75$. In Cleary (1998a), it

was shown that for normal mill operating conditions, the power draw was relatively insensitive to the choices of these two physical quantities. The same values are used throughout all the simulations in this paper in order to allow the effect of each changed condition to be independently evaluated.

The three key parts of our DEM implementation are the following.

- A search grid is occasionally used to construct a particle near-neighbour list. Using only particle pairs in this list reduces the force calculation to an $O(N)$ operation, where N is the total number of particles. Using such methods allows very large problems to be solved. Typically, problems with up to 100,000 particles are now easily soluble on single processor workstations.
- The collisional forces for each collision are estimated using the spring-dashpot model for each pair of particles in the near-neighbour list.
- All the collisional and other forces acting on the particles are summed and the resulting equations of motion are integrated:

$$\dot{x}_i = u_i, \quad \dot{u}_i = \sum_j F_{ij} + \mathbf{g}, \quad (3)$$

$$\dot{\theta}_i = \omega, \quad \dot{\omega}_i = \sum_j M_{ij}, \quad (4)$$

where x_i , u_i and F_{ij} are the position, velocity and collisional forces on particle i , and θ_i and ω_i are the particle orientation and spin produced by the moments M_{ij} and $\mathbf{g}$ is the gravity vector. The integration scheme is a second order predictor–corrector. Between 15 and 25 timesteps are required to accurately integrate each collision. This necessitates very small timesteps (typically 10^{-4} to 10^{-6} s depending on the controlling length and time scales of each problem). If the integration is accurate then the coefficient of restitution from which the damping coefficient was chosen should be recovered.

Quantitative predictions of wear rates and distributions, collision forces, dynamic loads on boundaries, power consumption, torques and flow rates, sampling statistics, mixing and segregation rates and many other quantities can be made from the information available in DEM simulations. For more details on the simulation method and on the data analysis, see Cleary (1998c). In this paper, we concentrate on predictions of power draw.

3. Calculation of torque and power

Every particle collision with the mill liner produces a torque on the mill. For each timestep these individual torques are summed to give the instantaneous torque on the mill. Only particles in contact with the mill contribute. If, for example, particles are in free flight, then they do not contribute to the torque until they strike the liner at the bottom of their flight. The product of the total mill torque with the angular mill speed (in rad/s) gives the instantaneous power consumption required to maintain the mill moving at the specified speed against this applied torque.

An alternative way of calculating the power draw is to sum the energy dissipation in each timestep for all the energy dissipation mechanisms for all particle collisions. The

energy dissipation in the normal direction is the product of the normal dashpot damping force with the normal approach speed. In the tangential direction, it is the product of the tangential speed of the particle contact with the tangential dashpot force if the contact is not sliding, or the product of the Coulomb friction and the tangential contact speed if the contact is sliding. This measure of power is much more difficult to obtain accurately because it is made up of many small contributions distributed over many timesteps. But, if the integration of the collisions is accurate then the average power calculated in this way should equal the power predicted from the torque. Since the torque-based power is more accurate we use this for all the results presented here.

The instantaneous power consumption is highly variable due to impulsive nature of the interactions. Averaging the power over several seconds gives a relatively invariant measure of the power draw that can be meaningfully compared to laboratory and field measurements. The choice of the averaging period is important because it determines the variability and reproducibility of the average. In principle, the longer the averaging period is, the less variability is present in the average. Unfortunately, whenever there is range of particle sizes present in the charge, then it can and does segregate rapidly. The amount of segregation increases the power consumption (Cleary, 1998a) and it increases strongly with rotation rate, leading to rises in power consumption as high as 15–20% for speeds of 105–110% critical. To make valid comparisons between different mill speeds and between cases with different particle distributions, the averaging period must be restricted so that the averaging is completed before any substantive segregation has occurred. The averaging period is therefore chosen to be between 3 and 10 s for almost all cases presented here. The first 3 s are omitted since this is the approximate time required for the slower rotating mills to reach equilibrium. This averaging period produces several hundred instantaneous power readings, which are calculated every 1000 timesteps. From a purely statistical perspective, this means that the errors in the power are less than $\pm 2\%$.

The length of the averaging period also depends on the number of particles in the simulation. When rocks are modelled as well as the balls, then the above averaging period is a satisfactory compromise between having enough data to get good statistical averages, but short enough to avoid structural changes in the flow due to segregation. When only balls are modelled, then much longer averaging periods are required to compensate for the increased fluctuations produced by the much smaller numbers of particles. Averaging periods of between 180 and 240 s are then required to give stable averages. Shorter averaging periods produce significant statistical variations of the predicted power vs. mill speed curve that do not have a physical basis. In general, a properly averaged power vs. speed curve should be reasonably smooth. DEM simulation results that demonstrate large irregular fluctuations should be treated with suspicion.

4. The ‘standard’ mill configuration

The ‘standard’ ball mill used throughout this study has an internal shell diameter of 5 m. There are 23 lifter bars spaced circumferentially around the mill shell. The standard lifter is symmetric with an 85° face angle and is 210 mm wide at the top and has a

height of 200 mm above the liner. The wear plates between the lifters protect the mill shell and reduce the internal diameter to 4.8 m. The ball mill rotates clockwise at various constant fractions N of the critical speed of 19.5 rpm, at which the particles begin to centrifuge.

The charge consists of both steel balls and rocks with a range of particle size distributions (PSD). The default charge consists of steel balls with diameters uniformly distributed between 50 and 100 mm and rocks with diameters uniformly distributed between 5 and 50 mm. The rock frequency is 10 times higher than the ball frequency, leading to a charge consisting of around 45% (by volume) of rock and 55% (by volume) of balls. The ball charge fraction actually varies between 53% and 58% due to the inherent randomness of the particle generation process. This charge composition is used for most of the power predictions here. Fill levels of 40% and 50% are used for most of the comparisons in this paper.

The standard configuration contains 5265 particles when the fill level is 40% and requires a computation time of 1.14 h/s of simulation on a 500 MHz alpha XP1000 workstation when using a conservative timestep on 9.04×10^{-6} . This is about 4.5 h per mill revolution.

5. Effect of rotation rate for the standard ball mill

5.1. Charge behaviour

Fig. 1 shows typical charge shapes predicted for our ‘standard’ 5 m ball mill and charge (described above) filled to 40% (by volume) for four rotation rates that span the typical range of operational speeds. For such a charge with a realistic size range, the free surface of the charge is well defined at low and intermediate rotation rates. For $N = 60\%$, material moves with the mill rotation around the bottom of the charge and up the left side of the mill to the shoulder. Here the net force on the particle declines to zero and the charge fluidises. Most of the material then flows down the steep free surface into the horizontal toe region. This is generally described as the cascading part of the flow. A small amount of material in the shoulder region is trapped between the lifters and is ballistically ejected into the mill cavity. This material drops onto the charge in the toe region and sometimes onto the unprotected liner of the mill. This part of the flow is generally termed as cataracting.

The particles are shaded according to their speed with light grey being low speed and dark grey being high. The fastest particles are the cataracting ones as they impact on the toe. The next fastest group of particles is the cascading or avalanching ones making up the free surface. They reach a peak speed just below the center of the mill and slow before arrival at the toe. The next fastest group of particles are the ones rotating with the mill near the shell. There is a semi-circular band (indicated by the light grey) of particles that move quite slowly. At the very center of this band is the center of the circulation pattern for which the particle speed is zero. Changes in the shading indicate shear that leads to particle size reduction by both abrasion and attrition. Such shear regions are clearly visible between the faster moving particles around the outside of the mill and the

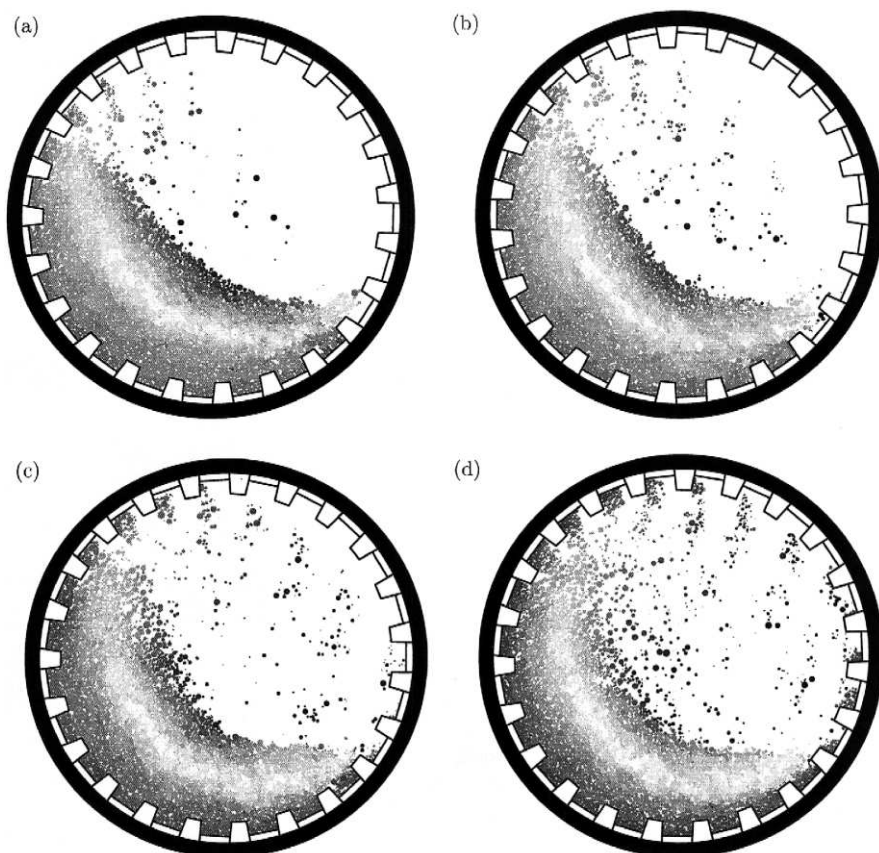


Fig. 1. Charge shape predicted for our 'standard' ball mill filled to $V = 40\%$ with rocks (5–50 mm) and steel balls (50–100 mm) for different rotation rates: (a) $N = 60\%$, (b) $N = 70\%$, (c) $N = 80\%$ and (d) $N = 90\%$. The particles are shaded according to their speed.

slowly moving interior particles and also between the cascading surface particles and the slow moving interior. The largest amounts of shear (indicated by the sharpest changes in grey scale) occur in toe region where material abruptly changes its motion from cascading down the surface to being pulled under the bulk of the charge by the lifters and swept to the left.

At $N = 70\%$, (Fig. 1b), the shoulder position moves slightly higher and the toe moves slightly lower. The inflection point on the free surface moves much closer to the center of the mill as the free surface profile becomes almost bi-linear. The amount of cataracting material has increased significantly and it possesses a clear banded structure. Each band is slightly curved, is almost vertical but tipped to the right a bit at the top and leads to the front face of one of the lifters above. Each lifter leaving the shoulder region acts somewhat like a filled bucket. As the lifter rises and its angle changes, the particles trapped inside gradually pour out with the last particles departing as the lifter reaches vertical. Each band corresponds to the material that poured from the space in front of the

lifter to which it leads. The impacts of cataracting material with the charge and mill liner in the toe region are therefore not continuous in time, but are bunched together and produce a cyclic variation in the impact forces in this region. For some rotation rates, spectral analysis of the instantaneous power draw detects a signal with a frequency given by the product of the mill frequency and the number of lifters that corresponds to the one that such cyclic impacts should produce. The distribution of particle speeds, and therefore shear, within the charge is little changed from the lower speed. The relative speed of the particles moving around the shell with the lifters compared to the speed of the cascading free surface particles has increased slightly.

Similar changes continue to occur when the mill speed is increased to $N = 80\%$ (see Fig. 1c). The amount of cataracting material has again increased strongly and their trajectories take particles close to the lifters above. The banded structure of these particles is more clearly visible. The shoulder again moves slightly higher and the toe marginally lower. The speed of the particles moving with the lifters and shell at the bottom again increases relative to the speed of the cascading particles. The semi-circular band of slow moving particles has become better-defined as the shear to either side becomes more concentrated. Near the shoulder region, the centrifugal force causes the free surface to become less sharply defined as this balances the gravitational force.

By $N = 90\%$, (Fig. 1d), the outermost cataracting particles barely move away from the shell and collide with the backs of the lifters slightly above the centerline of the mill. This would produce high wear on the lifters and is very undesirable. The fluidisation of the upper cascading free surface by the centrifugal force is now strong and the concentration of the shear into two semi-circular bands from the toe to the shoulder continues. For higher rotation rates particles become permanently trapped between the lifters and are then centrifuged for $N > 100\%$. The depth of the centrifuged layer increases with the mill speed N .

5.2. Variation of torque and power with mill speed

Fig. 2 shows the variation of the average torque and power with mill speed for the ball mill described above with a fill level of 50%. The torque increases slowly until the peak is attained around $N = 80\%$. As the amount of cataracting increases, the balance of the charge improves and the torque required to maintain the asymmetric charge position declines steadily. At $N = 100\%$, the charge begins to centrifuge and the rate of decline of the torque with speed increases.

The power, which is given by the product of this torque curve with the mill speed, has a much rounder peak centered on $N = 100\%$ before dropping sharply for higher rotation rates. For a mill operating at a typical 80% of critical, the power predicted by the model is 3×10^5 W/m of mill length. For a typical 7-m-long mill, the power required is therefore predicted to be 2.1 MW. This is consistent with a peak motor power rating of 3.3 MW that would commonly be used for a mill of this size.

Field data is very difficult to obtain for comparison with the DEM results so we instead compare with some of laboratory observations. Liddell and Moys (1988) used a 0.545 m mill (with smaller and more rounded lifters than the ones used here) and found that the torque peaked at 72% of critical speed for the 50% fill level used in the previous

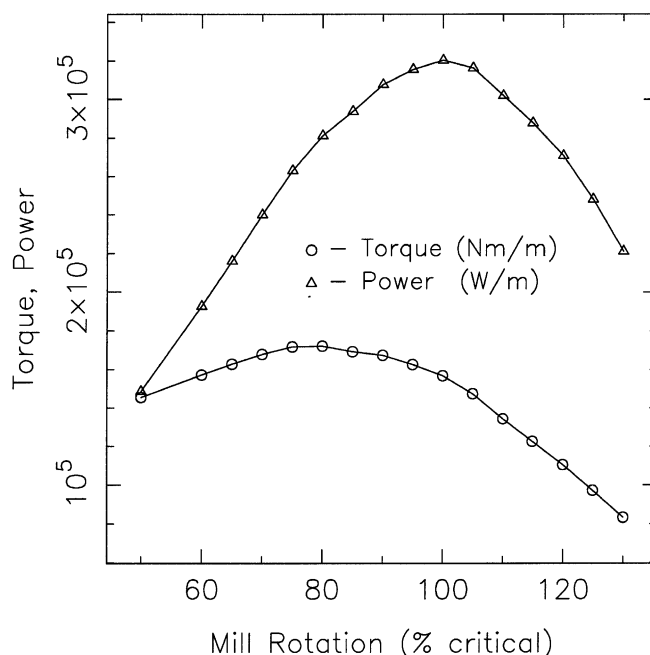


Fig. 2. Average torque and power as a function of mill speed N for the standard ball mill filled to 50% by volume. This is the base case for the following comparisons.

figure. This is moderately lower than the 80% speed predicted here by the DEM. Similarly, Liddell and Moys (1988) found the peak power occurred at $N = 82\%$ for this fill level, compared to 100% for the DEM prediction. This quantitative discrepancy arises because of differences in the two mill liners (the experimental lifters were smaller and more rounded and therefore less able to lift the charge) and from some of the simplifying assumptions made in the DEM model. It is therefore very important to ascertain the effects of various modelling assumptions on mill power predictions and to incrementally improve the DEM model so that these predictions become quantitatively better. For example, it will be shown later in this paper that the inclusion of rock shape is able to move the peak power from around 110% critical down to around 95% compared to 88% for these experiments. The inclusion of particle shape is able to explain most of this difference and is critically important for improving the quantitative accuracy of DEM mill predictions. Three-dimensional effects and the inclusion of the pulp slurry are also expected to influence this.

6. The effect of mill fill level

6.1. Charge shape and flow behaviour

Fig. 3 shows the changes in charge structure with decreasing fill level for our standard mill. When decreasing the fill level from 50% to 40%, the main change in the

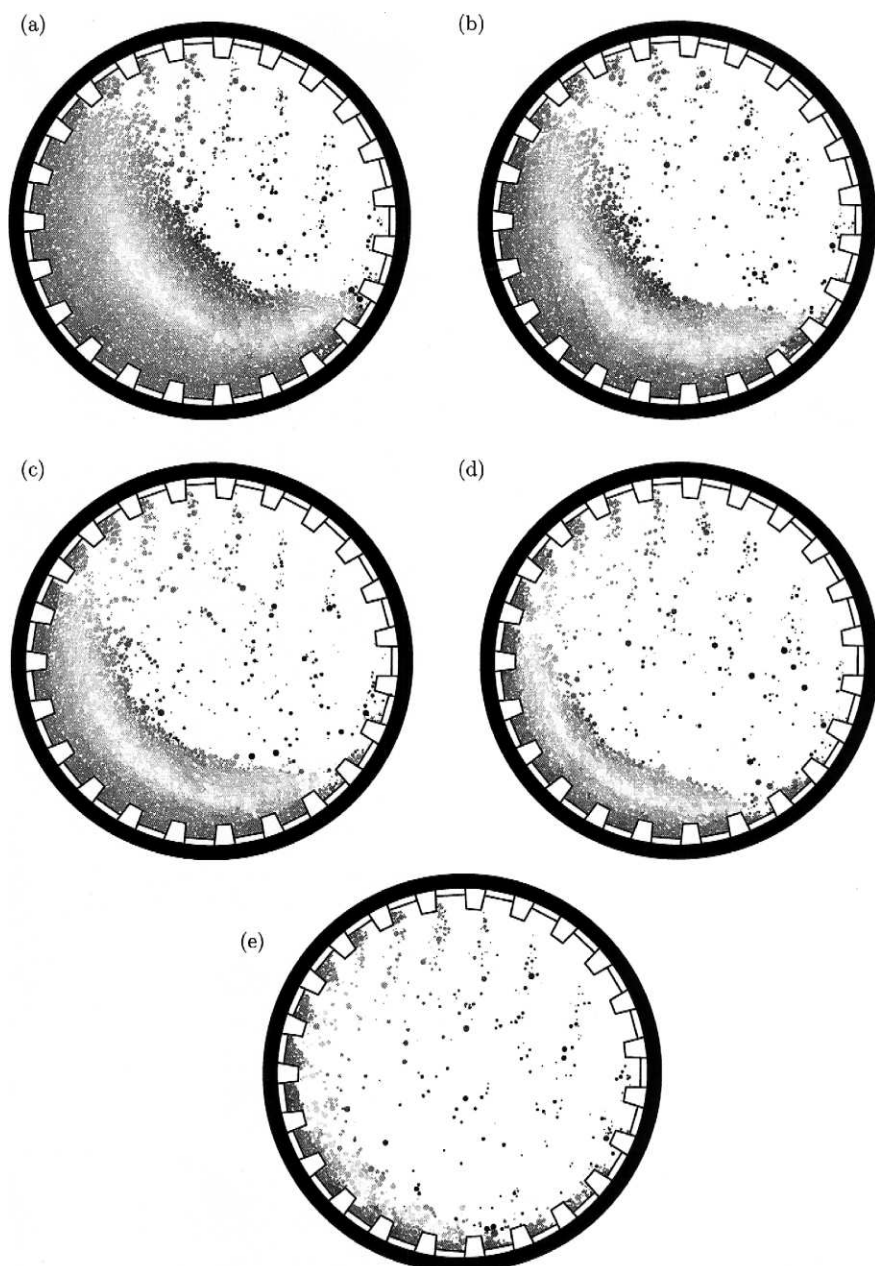


Fig. 3. Charge shape predicted for a ball mill rotating at 80% of critical speed and filled with rocks and balls with differing fill levels: (a) $V = 50\%$, (b) $V = 40\%$, (c) $V = 30\%$, (d) $V = 20\%$ and (e) $V = 10\%$. The particles are shaded according to their speed.

charge structure is the lowering of the cascading free surface. There is also a marginal increase in the amount cataracting material. Reducing the fill level from 40% to 30% again lowers the free surface. The shoulder and toe locations retreat only slightly so that the smaller amount of charge is spread out more thinly over only a slightly smaller fraction of the mill circumference. This behaviour continues with each subsequent reduction of the fill level, with the COM of the remaining charge approaching a shell of the mill. The trends observed with decreasing fill level can be summarised as follows.

- The free surface profile changes from being bi-linear to being increasingly curved.
- The shoulder position moves slowly lower and the toe moves slowly to the left, so the proportion of the mill circumference covered by the charge decreases much more slowly than the charge volume, leading to: (1) A charge mass that is progressively thinner. (2) The particle velocity near the shell remains that of the mill shell and the velocity of the cascading particles decreases only gradually with the gradual lowering of the shoulder. (3) The combination of decreasing charge depth with relatively constant velocity difference across the charge leads to steadily increasing shear rates within the charge and should therefore lead to more efficient grinding. The increase in the shear rates can be seen in Fig. 3 by the increasing gradients in the shading of the particles. In frame d, the fast moving particles at the bottom and top of the charge are very close to the curved central region of slow moving or stationary particles.
- For fill levels of $< 10\%$, there are almost no cascading particles. The charge depth is barely deeper than the depth of the lifter and almost all the particles move in the cataracting regime.

6.2. Power draw

Fig. 4a shows the power draw as a function of mill speed for fill levels between 10% and 50% for our standard mill. This clearly shows that a decrease in the fill level leads to a decrease in the power consumption. This general lowering of the power curve is expected, since all the material in the mill contributes to power consumption to one degree or another, so any reduction in fill level leads to reduced power consumption.

The torque required to support the charge in an off center position is essentially the product of the weight of the charge and the horizontal distance between the COM of the charge and the center of the mill. The power consumed is the product of this torque and the rotation rate. When the fill level is reduced from 50% to 40% the COM of the charge moves only marginally further to the left from its position somewhat below and to left of the center of the mill. This is because the material removed in lowering the fill level from 50% to 40% is very close to the center of the mill and has made very little contribution to the charge being off center and therefore very little contribution to the overall power consumption.

Comparing the material removed when lowering the fill level from 40% to 30% with that removed when going from 50% to 40%, we find that horizontal distance of the COM of this removed material from the center of the mill is more than double the size for the first removal. The amount of material removed is the same in both cases, so the more than doubling of the moment arm of the removed material more than doubles the reduction in power consumption. This behaviour continues with each subsequent

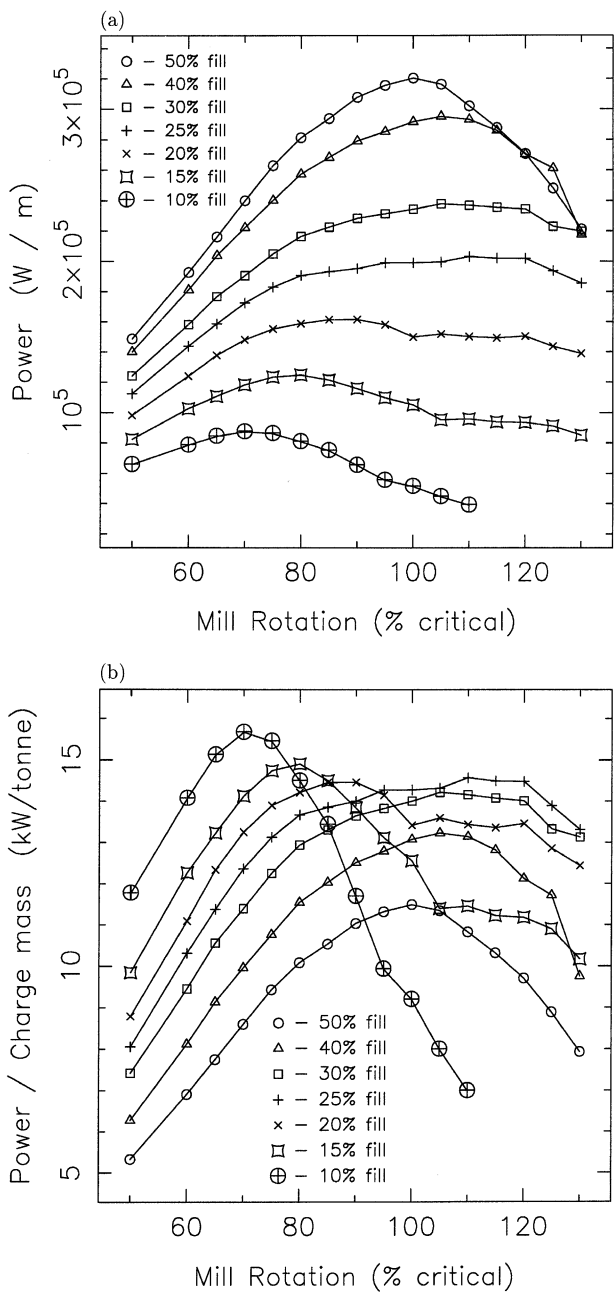


Fig. 4. Effect of mill fill level on power draw as a function of mill speed, (a) power draw and (b) power consumption/tonne.

reduction of the fill level, with the COM of the remaining charge approaching a shell of the mill in increasingly large increments. Each removal therefore makes a proportionally larger reduction in the power consumption. These changes are reflected in the relative movement of the power curve in Fig. 4a.

Of equal importance is the change in the structure of the power curve. This is best analysed by considering the power consumption per tonne of charge (known either as power density or specific power draw). It represents the rate at which energy is supplied to each unit of charge and controls the intensity of local grinding activity. Its variation with mill fill level is shown in Fig. 4b. When the fill level declines from 50% to 40% specific power consumption increases strongly (by between 15% and 20%). So although the total power consumed declines, the amount of power consumed per unit mass actually increases sharply. This suggests that more efficient grinding will occur for a lower total energy consumption at a 40% fill level rather than one jammed to near or above capacity at 50%. This is consistent with operational experience.

Decreasing the fill level to 30% again increases the specific power consumption, (but by a smaller percentage than that of the last increase). The peak specific power consumption remains around 105% but continues to broaden. For the 25% fill level the power curve is again translated higher by a smaller increment and again the peak broadens. This curve rises strongly and essentially linearly for speeds up to $N = 80\%$ at which point the gradient of the power curve drops abruptly. The specific power continues to rise much more slowly and essentially linearly to a peak at $N = 110\%$ after which it falls off rapidly as the charge centrifuges.

For fill levels between 25% and 20%, there is a strong qualitative change in the behaviour of the mill with the power curve developing two peaks. The dominant one moves to a significantly lower rotation rate between $N = 85\%$ and 90% and a broad flat minor peak appears between $N = 105\%$ and 120% . None of the mill conditions or parameters previously examined had had such an effect in changing the structure of the power consumption. Previously, the shape of the power curve and the location of its peak have been found to be insensitive to material properties of the charge the charge particle size distribution (Cleary, 1998a). This suggests that for speeds of $N \geq 93\%$ the mill efficiency may be higher for a 25% fill level, whereas for lower speeds it may be more efficient at the 20% fill level.

Decreasing the fill level to 15% continues this pattern of change. The dominant peak moves down to 80% critical and again increases in magnitude. The point of inflection in the curve moves from $N = 100\%$ to 105% and the minor peak moves to higher speeds and declines rapidly in magnitude. Decreasing the fill level to 10% continues these trends, with the peak in the specific power curve again increasing in magnitude and moving down to a speed of $N = 70\%$ and the power consumption at higher speeds collapsing. Moys (1999) reported that for a charge consisting of a single particle (the limit of decreasing fill level) that the peak power consumption occurs at a speed of $N = 50\%$. This is consistent with the trends identified using these DEM simulations.

In order to understand the effect of the changing of the fill level for a given mill operating at a single fixed speed, we re-plot (in Fig. 5a) the power as a function of the fill level for four common mill speeds. For each rotation rate the lowest power consumption occurs for the lowest fill level of 10%. The power then increases with

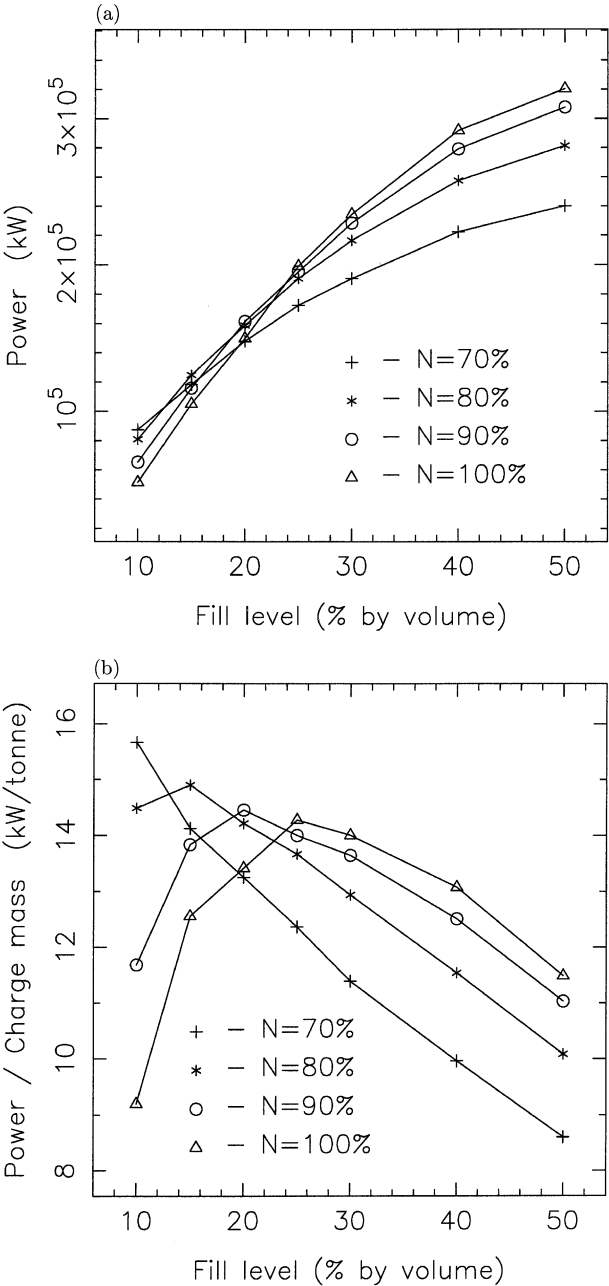


Fig. 5. Effect of mill fill level for four fixed mill rotation rates $N = 70\%$, 80% , 90% and 100% on (a) power consumption and (b) specific power consumption.

increasing fill level, but with steadily decreasing gradients. This occurs because each new mass added to the mill charge is physically closer to the center of the mill and therefore contributes progressively less torque and power than did the last addition of mass. Of particular interest is the observation that the gradients of these curves increase with the speed of the mill. The curve is much flatter at $N = 70\%$ than for the higher speeds. This suggests that power draw becomes more sensitive to the fill level as the mill speed increases. Also, at low fill levels, the highest power is used for the lowest speed, whereas at high fill levels the highest power is used for the highest speed.

Fig. 5b shows the corresponding specific power draw curves. For a mill speed of $N = 70\%$, the highest power intensity occurs for the lowest fill level of 10%. The power intensity then decreases almost linearly with increasing the fill level. For a mill speed of $N = 80\%$, the highest intensity to give occurs at an intermediate fill level of 15%. The intensity again decreases almost linearly as the fill level increases above this. The intensity is also lower for a 10% fill level, since for this speed much of this very small charge is almost being centrifuged by the lifters. For a speed of 90%, the power intensity peaks for a 20% fill level. For a mill speed of $N = 100\%$, the peak intensity occurs at a fill level of 25% and declines non-linearly with increasing fill level. The increasingly thick centrifuged layer causes the intensity to decline even more for the lower fill levels.

7. Effect of lifter shape

Sacrificial liners are used because of the high wear rates occurring within the mills. This wear occurs most strongly on the protruding lifter bars, reducing the lifter heights and their face angles over time. In order to evaluate the effect of the lifter life cycle on mill performance, a series of simulations were run for lifters with progressively shallower face angles. The width of the lifter at the base was kept constant while the height of the lifter and its width at the top were suitably reduced to maintain plausible aspect ratios. The wear on the lifters was assumed to be symmetric, as would be the case for mills that are periodically reversed. Table 1 shows the geometric details of the four lifter sets corresponding to lifter face angles of 85° (newly installed), 60° (moderately worn), 45° (well worn) and 22.5° (just before replacement).

Table 1
Geometric details of the four lifters used to represent the wear-induced changes in lifter shape throughout the lifter life cycle

Lifter face angle (°)	Lifter top width (mm)	Lifter height (mm)
85	200	200
60	110	142
45	76.25	102
22.5	76.25	52

Fig. 6 shows the charge behaviour in the mill for these four lifter shapes for a 40% fill level. The 85° face angle case (Fig. 6a) is the same as the ‘standard’ configuration (Fig. 1a), but with particles now shaded by size. The main characteristics of the charge are a bi-linear profile with the free surface inclined slightly down towards the toe and a moderate amount of cataracting material falling in bands through the upper part of the mill. When the lifter is moderately worn with a face angle of 60° (Fig. 6b), there are only subtle changes to the charge behaviour. The bi-linear free surface is largely unchanged, but the toe location has risen slightly. The main change is that the cataracting material is thrown on much lower trajectories by the shallower face angle of the lifters. Similar changes again occur for the more worn 45° face angle case in Fig. 6c. The shoulder location is a little lower and the toe is marginally higher. The amount of cataracting material is much reduced and even the highest trajectory material lands on the charge surface about half way between the mill center and the toe. When the lifter

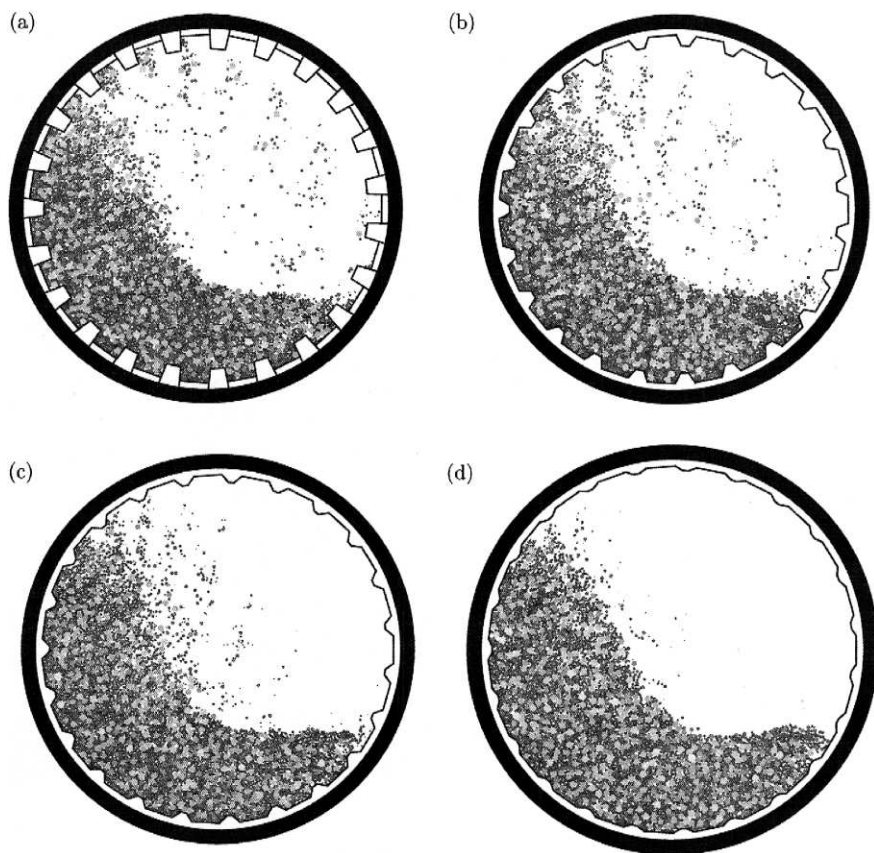


Fig. 6. Charge shape predicted for a ball mill rotating at 80% of critical and filled with rocks and balls for lifters with face angles: (a) 85°, (b) 60°, (c) 45° and (d) 22.5°.

has been reduced to a 22.5° face angle, there is almost no cataracting material at all and it barely moves beyond the avalanching surface of the charge. The toe location has moved noticeably higher and is now higher than the surface at the bottom of the avalanching region in the middle of the mill. The change in charge behaviour as the liner wears is gradual, but the cumulative effect is quite large.

Fig. 7a shows the specific power as a function of mill speed for the four lifter geometries. Each power curve is essentially parabolic in shape. At low speeds (e.g. $N = 50\%$), the highest power draw occurs for new 85° lifters and decreases slowly with each reduction in face angle. At high speeds, the opposite is true, with the highest specific power consumption occurring for the shallowest lifter and the lowest power draw occurring for the steepest lifter. This occurs because the steep lifters are able to trap more of the near centrifuging material. As the lifter face angle and the height declines then more of the charge remains mobile, flowing across increasingly gentle lifter undulations. More specifically, for speeds $N < 70\%$, the steepest (new) lifters produce the highest power draw. At this speed, the 60° lifter takes over as having the highest power draw. At a speed of $N = 82\%$, the 45° takes over having the highest power draw and finally for $N > 94\%$, the 22.5° has the highest specific power draw. This demonstrates that intensity of collisional energy dissipation changes significantly throughout the lifter life cycle.

Fig. 7b shows the specific power draw as a function of the declining lifter face angle for four common mill speeds.

- For a mill speed of 70%, the power consumption varies only slightly throughout the lifter evolution. The specific power is essentially unchanged from 85° to 60° and then declines gradually and linearly. The total decrease in energy supplied to the charge by worn lifters is only 4.7% lower than that produced by a new liner.
- For a mill speed of 80%, the specific power draw actually increases as the face angle wears from 85° to 60° , then declines slightly at 45° and more rapidly at 22.5° . This means that peak energy consumption does not occur for new liners when operating at this speed, but increases with some wear, peaking for a 60° face angle and then declining.
- For a mill speed of 90%, the lowest specific power consumption occurs with the new lifters. As the lifter wears, the power increases with the peak occurring for a face angle of 45° . The mill performance then drops off as the lifter wears further. The change in power consumption between the best and worst lifter shapes is now 8.4%.
- For a mill speed of 100%, the lowest specific power occurs for new lifters. Energy consumption increases as the lifters wear and is still increasing even at 22.5° , by which point it has increased by 16.5%.

The lifter face angle for which the peak power consumption is expected decreases with the mill speed. For high speed mills operating at between 90% and 100% (as is common in South Africa), the best face angles would appear (on this basis) to be between 22.5° and 45° . The amount of change in the power draw with decreasing face angles increases with mill speed. For a speed of 70%, the power is relatively insensitive to the lifter face angle, whereas for a speed of 100%, the power varies strongly throughout the lifter life cycle. This suggests that mill control will be more difficult as the speed increases.

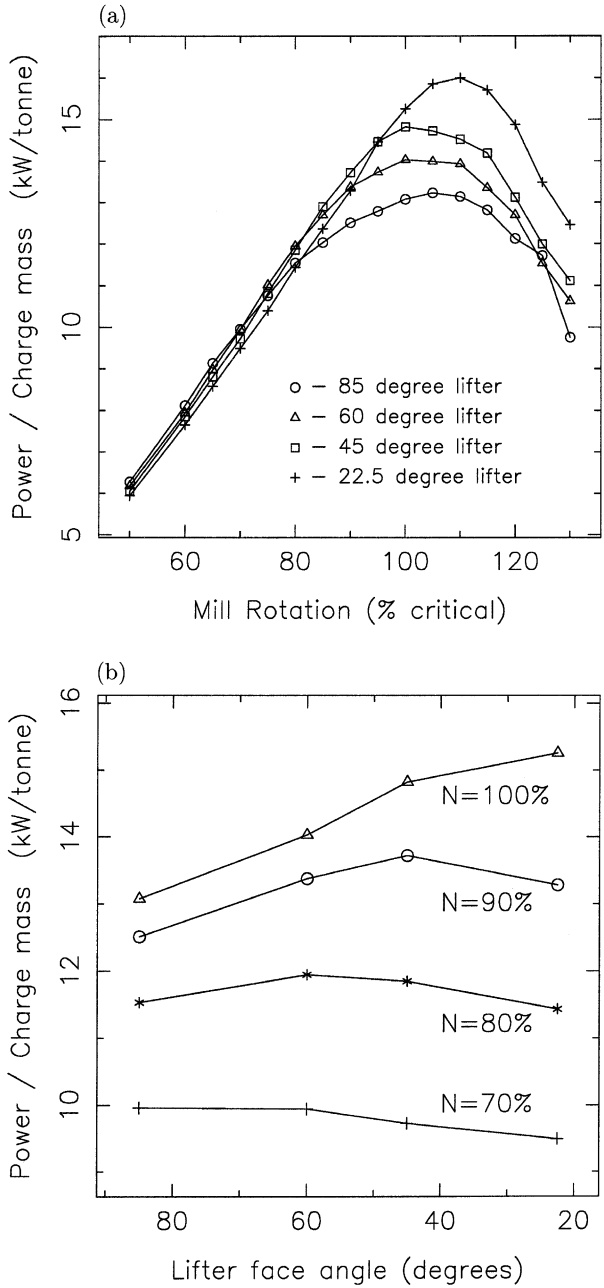


Fig. 7. (a) Dependence of the specific power consumption vs. mill speed curve on the lifter shape for face angles between 85° and 22.5°. The fill level here is 40%. (b) Change in specific power consumption with face lifter angle throughout the lifter life cycle for the rotation rates $N = 70\%$, 80% , 90% and 100% .

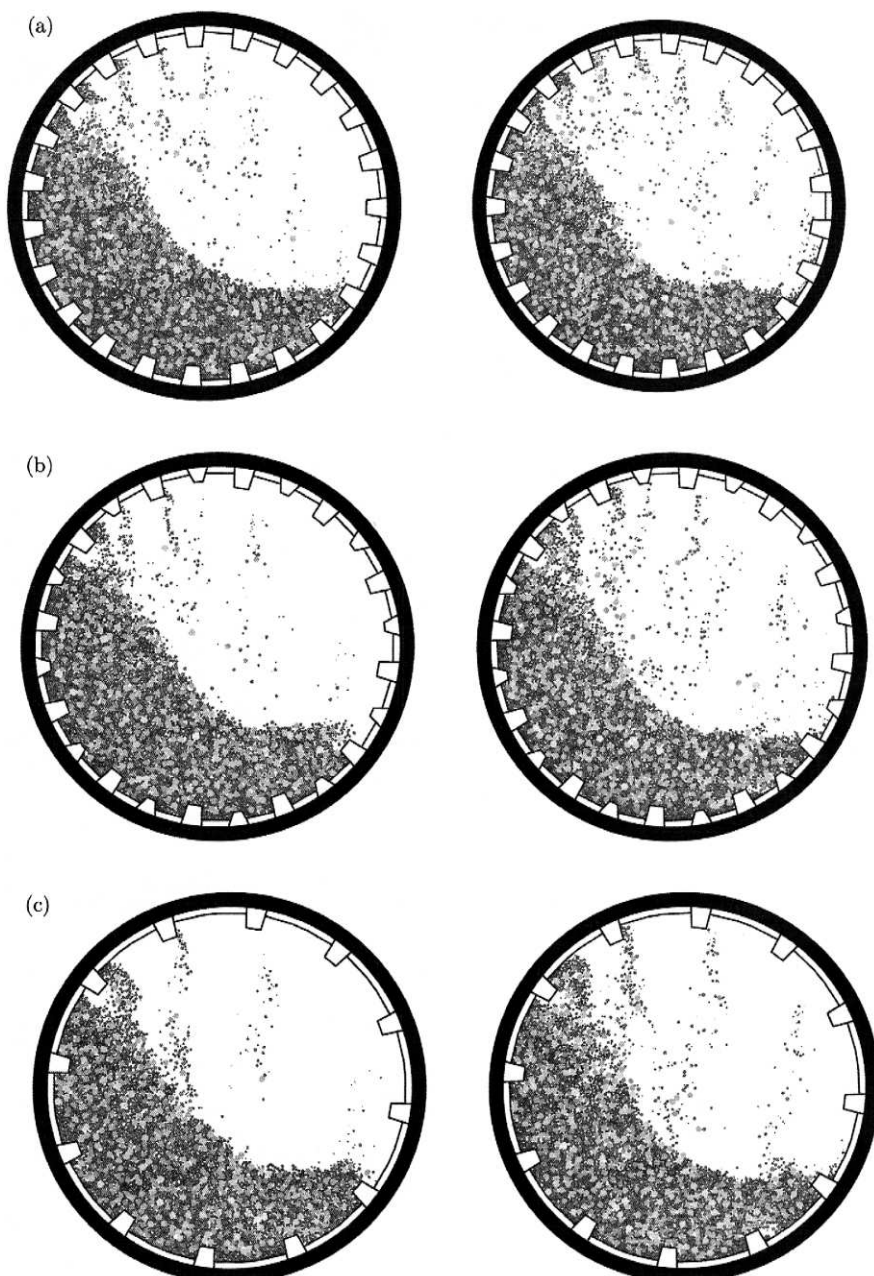


Fig. 8. Charge shape predicted for a ball mill with (a) a conventional hi-hi lifter pattern, (b) a hi-lo pattern and (c) with half the number of lifters (12). The mills in the left column have a speed of $N = 70\%$ and the mills on the right rotate at $N = 80\%$. The fill level is 40%.

8. Effect of lifter pattern

The radial distribution of lifters around the mill shell is also important. Here we compare the performance of three important lifter patterns.

- The standard hi–hi lifter pattern used throughout this paper.
- A hi–lo pattern consisting of alternating high and low lifters. This is achieved by changing only half the lifters at each reline, so that half are older and half are newer lifters at any particular time. This is modelled using 12 standard 85° lifters alternating with moderately worn 60° lifters.
- A hi–blank pattern where every second lifter is omitted, thereby substantially increasing the spacing between lifters.

Fig. 8 shows the charge behaviour for the three liner patterns for two mill speeds in the common operational range, 70% and 80% critical. Changing from a hi–hi to a hi–lo pattern produces only subtle changes to the charge distribution. The lower face angles on the low lifters means that the corresponding cataracting material is thrown on lower trajectories. This produces modestly less cataracting material and also smears its banded structure. The toe location also moves slightly higher. Removing the low lifters eliminates every second band of the cataracting material. The toe location moves slightly higher again and the bi-linear profile of the charge surface becomes much more rounded.

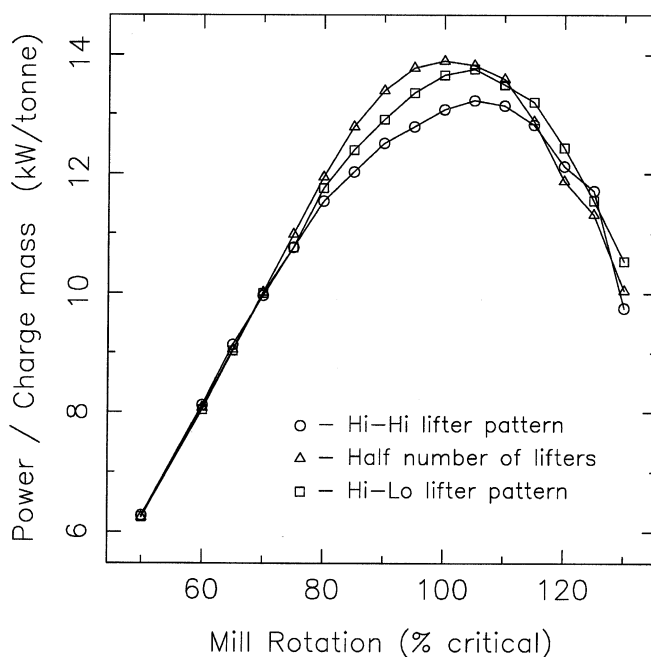


Fig. 9. Specific power consumption for three lifter bar patterns; hi–hi, hi–lo and with half the number of lifters.

The variation in the toe location with the lifter pattern is somewhat larger for the 80% mill speed than for the 70% speed.

Fig. 9 shows the corresponding predictions for the specific power consumption. For speeds up to and including 70% of critical, there is no dependence of the power on the lifter pattern. Above that, the power consumption becomes increasingly sensitive to lifter pattern. The hi-lo lifter configuration has consistently higher power draw than the standard pattern, until the centrifuged layer produced at very high speeds reaches the tops of the lifters ($N = 120\%$). The lifter pattern consisting of half the standard number of lifters produces an even greater increase in specific power for this range of mill speeds. This suggests that halving the number of lifters should, for higher speed mills, increase energy input into the charge.

9. Effect of ball and rock size distribution

One of the largest sources of variation in mill performance is in the feed material and the proportion of the charge that is made up of steel balls. There are several attributes of the rock distribution whose effect on mill performance need to be explored, both to assist in understanding mill behaviour better and in understanding the effects of various modelling assumptions on the DEM predictions. Changes in the material properties such as the friction coefficient and the coefficient of restitution have previously been shown to be relatively unimportant (Cleary, 1998a) and will not be examined here. Here we will concentrate on the effect of changes in PSD. From a modelling perspective, this is very important because the entire size range cannot be simulated and some degree of approximation is required.

9.1. Effect of the shape of the particle size distribution

9.1.1. Discrete vs. uniform continuous ball sizes

Fig. 10 shows the specific power draw as a function of mill speed for three different types of size distribution with a fill level of 50%:

1. The circles correspond to a charge consisting entirely of balls in three discrete sizes, 5% (by number) of 200 mm diameter, 47.5% of 100 mm and 47.5% of 75 mm.
2. The triangles correspond to a charge consisting entirely of balls with a continuous uniform distribution between 50 and 150 mm.
3. The squares correspond to a combination of a continuous ball distribution between 50 and 200 mm and rocks with sizes uniformly distributed between 10 and 50 mm.

For $N \leq 75\%$, there is negligible difference in the specific power draw for the first two ball only cases. Between 80% and 100% critical there is a small difference of 2% to 5% with the continuous ball size distribution being lower. For super-critical speeds, the

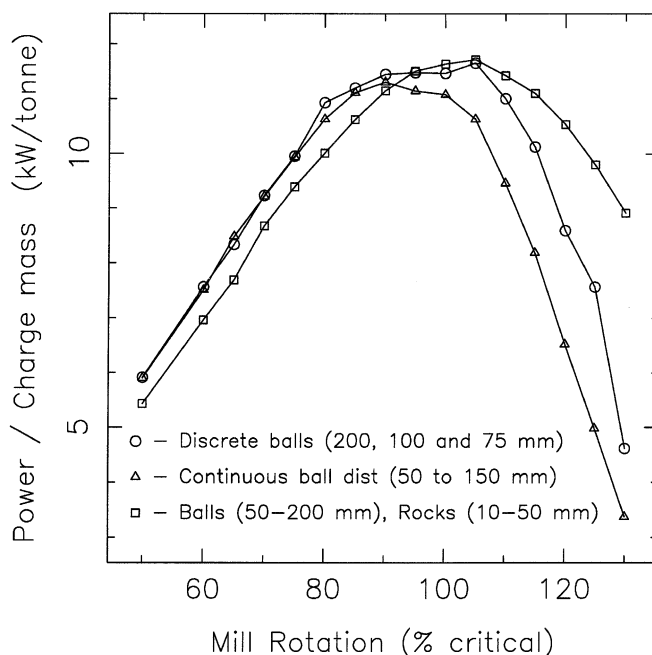


Fig. 10. The effect of the shape of the ball size distribution on the specific power consumption.

continuous ball case shows a significant reduction in power consumption of 20–25%. This demonstrates that the power draw is sensitive to even small changes in the ball distribution.

Comparing the ball only cases to the ball and rock case, we find that the inclusion of rocks leads consistently to a 0.5-kW/tonne lower specific power draw prediction, for speeds below 80%. This is a decrease of 5% to 8% and occurs in the speed range where many mills are operated. This results from a higher toe location in the rock and ball case producing a more balanced charge and therefore requiring a lower torque and power consumption to support it. At higher speeds the ball only curves have different shapes and different maxima, with the ball and rock specific power exceeding both the ball only cases at around 94% of critical speed. For higher speeds, the predictions diverge substantially, indicating that the omission of the rock component from such simulations is likely to lead to substantially erroneous power predictions.

9.1.2. Uniform number distribution of rock size vs. uniform mass distribution

In the majority of the simulations used in this paper, the particle distribution has been a binary mixture of balls and rocks each with a continuous uniform distribution between upper and lower limits. The ball size distribution in a real mill is generally somewhat controllable and depends on the replacement strategy used for the mill. The uniform size distribution is a plausible approximation for this and corresponds to a strategy of regularly adding new balls of the largest size to the charge and occasional removal of balls smaller than 50 mm (either deliberately or by breakage).

The choice of a uniform distribution for the rock sizes does not necessarily represent the actual size distributions occurring in real mills. In general, there are more particles in the smaller size ranges than in the large. When a large particle breaks, it produces many daughter fragments of smaller sizes. Partially counter-balancing this is the removal of some smaller particles at the discharge end of the mill. A plausible generic particle size distribution is one where there are equal masses of particles in each part of the size range. In two dimensions, this means that there are four times as many particles of a given size as there are of particles with twice that size. This much higher frequency of smaller particles substantially increases the computational problem size. Nevertheless, it is important to understand the effect that using different types of size distributions has on the charge behaviour and the power predictions.

Fig. 11 shows the charge distribution in the mill for the standard configuration with a fill level of $V = 40\%$ and a speed of $N = 80\%$ for the two different frequency distributions for the rock size. The particles are shaded by size. The regions between the lighter shaded balls are much darker for the mass weighted size distribution, demonstrating the reduction in the average size of the rock particles. The change in the size distribution has a number of observable effects.

- The cataracting streams falling from the spaces between the lifters are much more clearly defined because of the greater number of particles in each stream.
- The toe position is slightly higher and the shoulder position slightly lower. This shows that the charge is weaker when there are more smaller particles and the charge slope fails and flows at a slightly lower angle.
- The shape of the free surface of the charge is noticeably different for the mass weighted case. It is slightly lower at all points indicating that the charge packs slightly better and so occupies a smaller volume. In both cases, the surface is bi-linear with an inflection point slightly to the left of the mill center. For a uniform number density the cascading region is more dilated and the surface always slopes down from the inflection

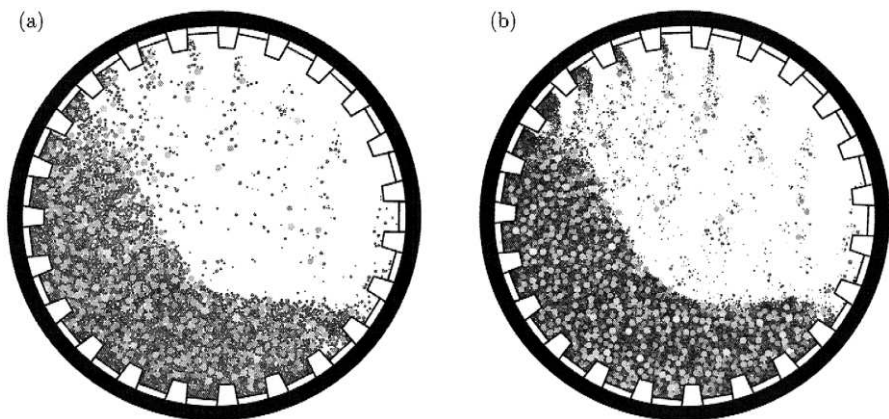


Fig. 11. Charge behaviour for a mill with a fill level of $V = 40\%$ and a speed of $N = 80\%$ for (a) uniform number frequency for the rock size distribution and (b) uniform mass frequency.

point to the toe. For the mass weighted case, this lower section is essentially horizontal. This indicates that the charge containing more fines is able to flow better and more of it is able to reach the toe causing it to move higher.

Fig. 12 shows the specific power consumption for these two types of rock distributions. The power curve for the mass weighted case is lower across the entire sub-critical range of speeds and the peak power has moved from $N = 110\%$ to 120% critical. The power is much higher at super-critical speeds than it was for the original size distribution. The lower power consumption at $N = 80\%$ is consistent with the changes observed above in the charge distribution. That is, the lowering of the shoulder and raising of the toe decreases slightly the amount by which the charge is out of balance and thereby reduces both the torque and the power draw.

The changes observed in the power curve when increasing the proportion of smaller rocks are a continuation of the same trends exhibited earlier. These changes can be interpreted in terms of the strength of the charge structure.

- For *sub-critical speeds* the main part of the charge occupies the region below the characteristic bi-linear free surface. If the charge structure is weakened, then the cascading slope fails more easily and has a lower dynamic angle of repose leading to a lower shoulder angle. The failed material is also more able to flow in the direction of the toe and causes the toe to move slightly higher. This leads to a reduction in the power consumption. If the charge structure is strengthened, then the opposite effects occur and the shoulder position moves higher and the toe lower leading to higher power consumption.

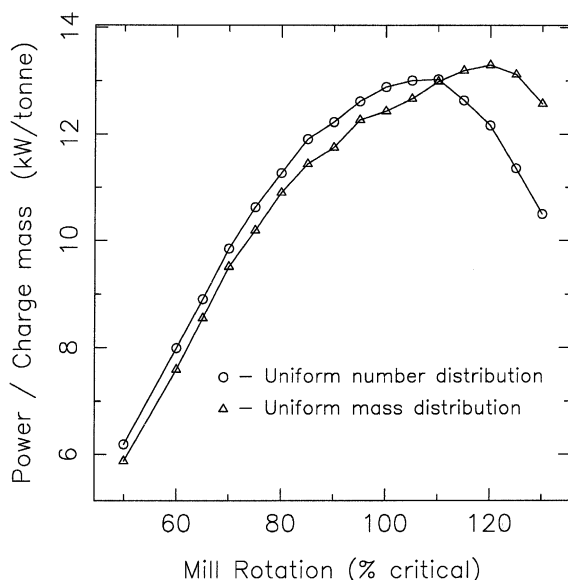


Fig. 12. The effect of the shape of the rock size distribution on the specific power consumption. The circles show the power prediction for a uniform number distribution for the particle size while the triangles show the power for a uniform mass distribution, which has substantially larger numbers of smaller particles.

- For *super-critical speeds* the particles are distributed in a centrifuged layer around the inside of the mills. If the charge structure is weakened, then the particles near the free surface of this centrifuged region become more mobile. They therefore dissipate more energy in collisions and are able to re-distribute themselves leading to the charge being more out of balance. This increases the power consumption. Conversely, if the charge structure is strengthened, then particles are trapped more efficiently and held immobile at the free surface leading to less motion and less energy dissipation and therefore lower power consumption.

The combination of these effects necessarily means the peak power consumption moves to higher speeds as the charge structure is weakened. The addition of the rock fraction to the ball only case weakens the charge and pushes the power curve down and to the right. Similarly, the addition of more fines ‘lubricates’ the flow leading to a weaker charge and a further reduction in power for sub-critical speeds and an increase for higher speeds. This demonstrates:

- the effect of increasing fines within the mill on the power draw, and
- the overall importance of having the correct structural strength for the charge in DEM simulations of mills in order to correctly predict the power draw.

9.2. Effect of the ball fraction

Previously, the ball fraction of the charge has been around 55% (by volume). This is a reasonable generic value, but the ball fraction is one of the charge attributes (along

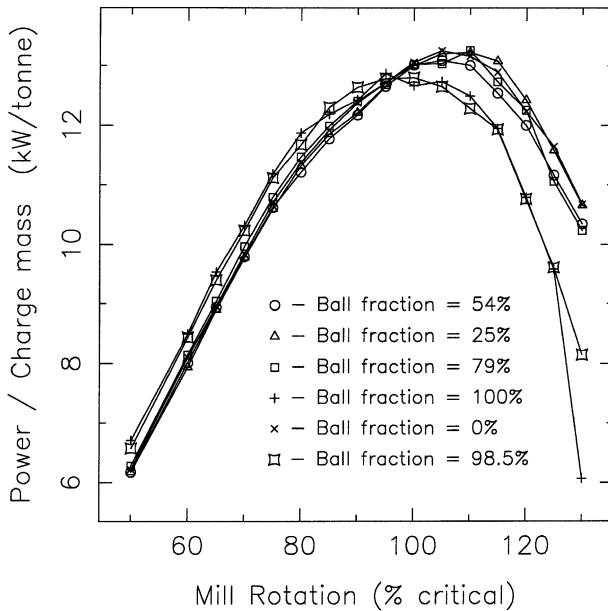


Fig. 13. Effect of ball fraction on the specific power consumption.

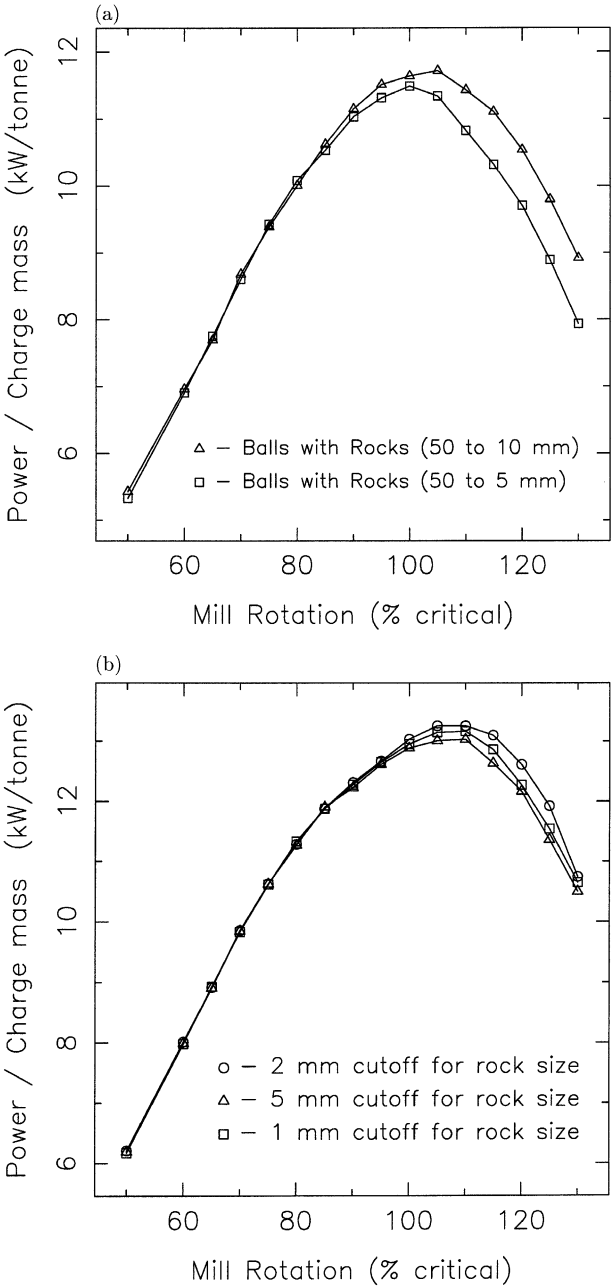


Fig. 14. Effect of the lower cutoff of the size distribution on the specific power consumption, (a) 5 vs. 10 mm for an AG mill, and (b) 1, 2 and 5 mm cutoffs for the rock size in a ball mill.

with ball size distribution) that can be varied by operators, so it is useful to understand its effect on mill behaviour. Fig. 13 shows the specific power draw for ball fractions varying from 0% (fully autogenous) to 100% (ball only or fully ground out). The specific power curves for all the cases with ball fractions $< 80\%$ are very close together. The differences are very small for all sub-critical mill speeds with larger but still modest differences at super-critical speeds. Cases with ball fractions as high as 80% behave essentially the same as the autogenous case. However, for cases with either no rocks or very few rocks, the interlocking of the balls becomes stronger, leading to an increase in specific power at lower speeds and a reduction at super-critical speeds. This change in behaviour occurs only for very high ball fractions between 80% and 98.5% and indicates that the rocks are a significant contributor to determining the shear strength of the charge even when the amount of rock is very low. This again demonstrates the danger of using ball only DEM models to predict power draw for cases where there are actually rocks present.

9.3. Effect of lower cutoff of rock size distribution

The computational cost of DEM simulation is proportional to the number of particles. The current limit for reasonable computational times is around 100,000 particles for a single processor workstation. It is therefore not possible (and unlikely to be possible for many years) to simulate the entire size range in a real mill. The particle size distribution must therefore be truncated at some specified diameter and the smaller particles either omitted or treated in some other manner. It is very important to understand the sensitivity of DEM predictions to this cutoff.

Fig. 14a shows the specific power consumption for two AG mills with lower particle size limits of 10 and 5 mm, respectively. For mill speeds $N \leq 80\%$, the specific power draw predictions are identical, but for speeds between 85% and 110% critical, there is a small difference of 2–3%. For higher speeds, the difference is moderately higher.

Next we consider the ball mill (rock and ball) cases with smaller cutoffs. The base case is our standard mill with a cutoff of 5 mm for the rock size. This is compared to cases that are same except for the cutoff being reduced to 2 and 1 mm, respectively. The specific power draw curves are shown in Fig. 14b. The predicted power is indistinguishable for the three cutoffs for the speeds up to $N \leq 85\%$. Very small differences occur for speeds up to 100% of critical speed. For the super-critical speeds there are some small differences up to 3.5% at $N = 115\%$. As the speed increases further, the charge increasingly centrifuges and the power differences decrease again. This suggests that power predictions are insensitive to lower cutoff sizes in the range of 1–5 mm for low and medium speeds and that 5 mm is a reasonable cutoff for higher speeds.

10. Effect of particle shape

So far, the particles have been modelled as disks, but particle shape has previously been shown to be important in a range of industrial granular flow applications, particularly in mixing during avalanche flows in rotating drums (Cleary et al., 1998;

Cleary, 2000) and hoppers (Cleary, 1999). The majority of rocks in mills are not really circular, particularly those resulting from the fracturing of hard rock ores. It is therefore important to understand the effect of variations in the rock shape on the power predictions. This is important because variation in shape distribution of the feed and its fracturing characteristics are likely to lead to variability in mill performance and because the complete shape distribution can never be measured and must be approximated for modelling.

As described in Section 2, we will model non-circular particles using super-quadrics with various amounts of blockiness and differing aspect ratios.

10.1. Effect of ball shape

The effect of non-circularity on both the flow and power draw is evaluated for the case where the mill contains only non-circular steel balls. We use the same configuration as described in Section 9.1 with three discrete ball sizes and allow the balls to have aspect ratios distributed between 1 and 2:1 and super-quadric powers between 2 and 5. The balls vary in shape from round to squarish.

Fig. 15 shows the ball charge behaviour with circular and non-circular balls for three mill speeds. At $N = 70\%$, subtle differences appear in the charge distribution. The free surface of the charge composed of non-circular balls is higher than for circular balls. This occurs because of the dilation of the cascading flow by the non-circular particles that require more space in which to rotate. The shoulder is also raised higher causing the cataracting stream to be thrown on a higher trajectory. Similar changes are observed at $N = 85\%$ with the dilation near the top of the cascading region being particularly strong. At $N = 95\%$, the charge distribution is quite different. For circular balls, the bulk of the charge lies in the lower left side of the mill with a modest amount of cataracting material passing through the middle. For non-circular particles, the charge is significantly dilated and is spread throughout a much larger area of the mill.

The specific power consumption for these two ball shapes is shown in Fig. 16. For $N \leq 75\%$ there is no change in the mill power draw. This occurs because the center of mass of the charge is little changed by the particle shape at these speeds. At intermediate speeds, the power draw for the non-circular balls falls off rapidly. This reflects the increasingly well spread out and dilated charge for the non-circular particles. At $N = 95\%$, the charge is much more evenly spread throughout the mill than for the circular case, leading to a less out of balance charge, a lower torque requirement and therefore lower power consumption. This difference in behaviour can again be related to the charge strength. Circular particle structures are much weaker than non-circular ones and yield much more easily. In this case, the slope of circular particle is able to fail and flow easily maintaining the charge in the lower left side. The non-circular particle structure is stronger, delaying slope failure so that particles are raised much higher and are thrown throughout the mill leading to a more even particle distribution and therefore lower torque and power draw.

The inclusion of particle shape has caused the peak power draw to move to much lower speeds, from 105% to 90% critical. This brings the DEM predictions much closer to experimental and field observations where the peak generally lies in range 80–90% of

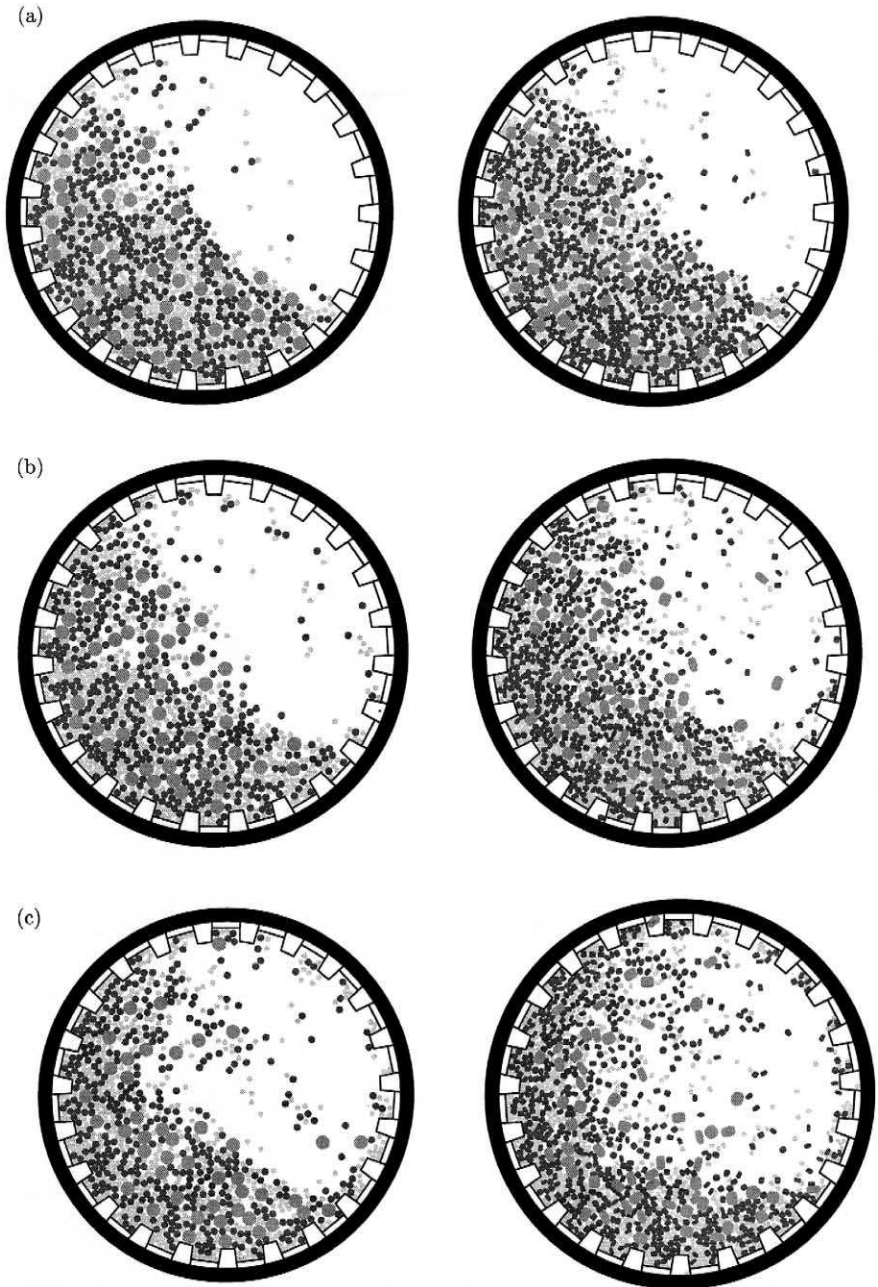


Fig. 15. Charge behaviour predicted for non-circular balls in a mill rotating at (a) $N = 70\%$, (b) $N = 85\%$ and (c) $N = 95\%$. The left column shows the flow with circular balls and the right column shows the flow at the same time using non-circular balls.

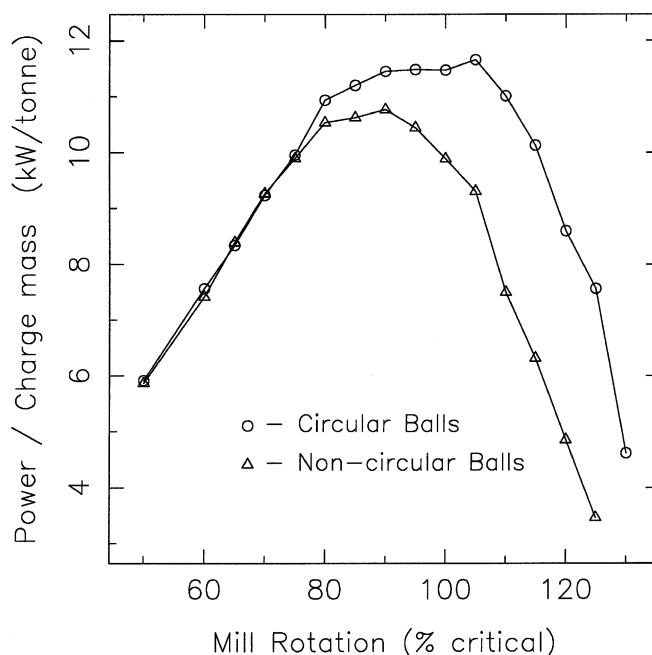


Fig. 16. Effect of the ball shape on the specific power consumption.

critical speed. The remaining differences are due to the use of two dimensions, since particles are expected to interlock better in 3D leading to further increases in charge strength and further movement of the peak power to lower speeds.

10.2. Effect of rock shape

Next we consider non-circular rocks in a charge with circular steel balls. The rock aspect ratio varies from 0.4 to 1.0 and the blockiness from 2.1 to 5.0. Two charge types are considered: (1) The particle shape is independent of particle size. (2) The particle shape varies with the size of the particles. The largest particles are given the largest variations in shape and the range of shapes decreases linearly with the particle diameter, so that they become progressively rounder as they become smaller. This attempts to mimic the real changes in particle shape that occur in mills due to the rounding produced by chipping of corners and attrition.

Fig. 17 shows close ups of the charge structure for the two sets of rock shapes. Fig. 18 shows the specific power consumption for the two non-circular rock cases and for the standard circular rock case for a fill level of 40%. At sub-critical speeds, the circular rocks have the lowest power consumption, while at super-critical speeds they have the highest power consumption. The power consumption for non-circular rocks (with shapes becoming rounder with size) is modestly higher than the circular particle power for all sub-critical speeds and quite a bit lower for the super-critical speeds. The non-circular

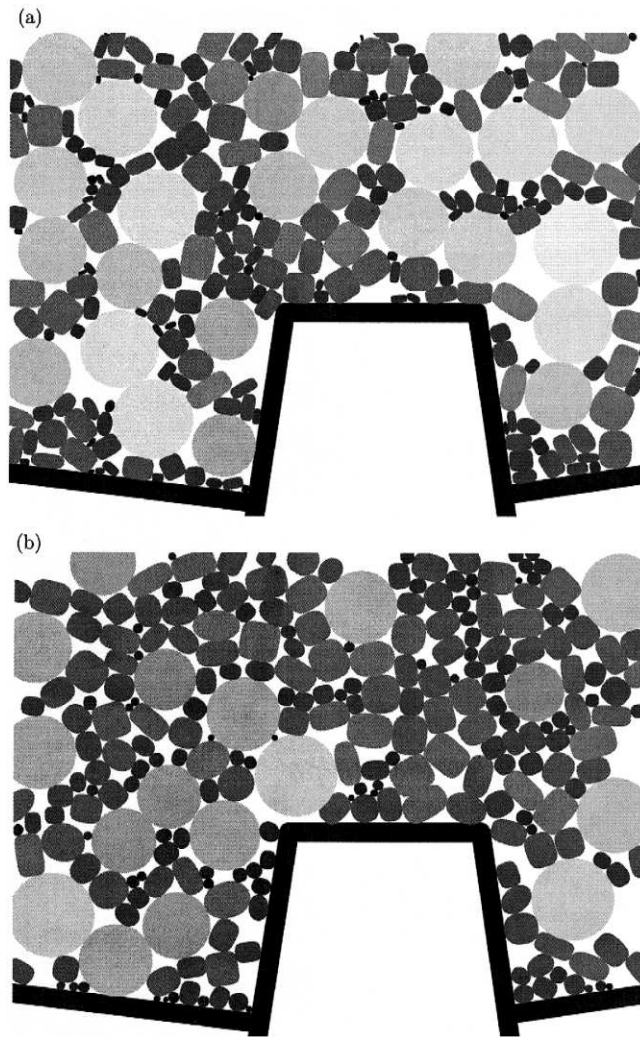


Fig. 17. Close up of particles with non-circular shape when (a) shape is independent of particle size and (b) when the particles become rounder as they become smaller. The particles are shaded according to their size.

rocks (with shape independent of size) have the highest power consumption at low speeds and the lowest power draw at high speeds. The speed at which the peak power consumption occurs varies from 110% critical to 100% and then down to 95% as the particle shape variation increases. For this 40% fill level, Liddell and Moys (1988) experimentally found the peak power at $N = 88\%$. These results are now very close, particularly when considering the smaller rounder lifters used in these experiments. This demonstrates the critical importance of particle shape in DEM modelling.

All these variations in the power curve can again be explained in terms of the shear strength of the charge structure. The non-circular rocks with independent shapes have

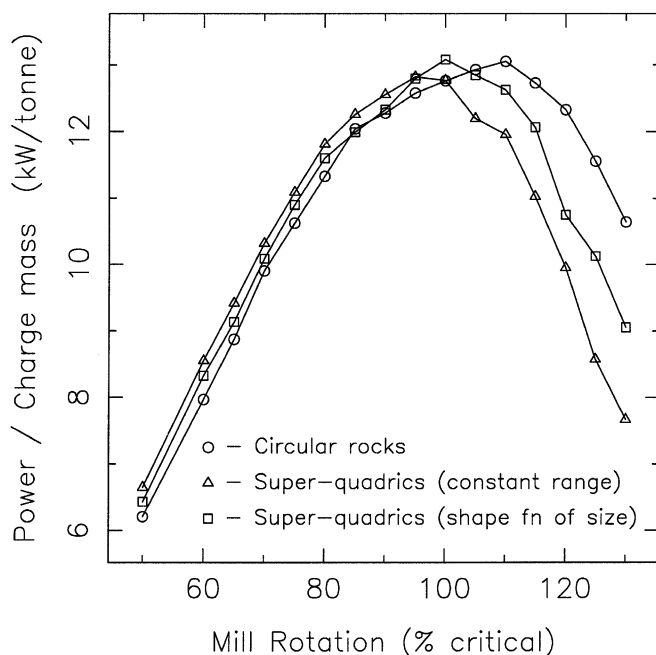


Fig. 18. Effect of the rock shape distribution on the specific power consumption.

the largest variation from circular allowing the rocks to lock together the most, producing the strongest structure. This leads to the highest dynamic angle of repose, the highest shoulder position and therefore the highest sub-critical power draw. It also leads to the highest locking of the particles in the centrifuged layer and therefore the lowest power draw for the super-critical speeds. The non-circular particles with shape dependent on size lock together better than the circular rocks but not as well as the other non-circular case giving an in-between power curve.

11. Conclusion

Two-dimensional DEM modelling has been used to systematically study the effects of a wide range of mill and particle properties on the charge shape and the power draw of a ball mill. This is important for helping to understand the effects of operational, liner and feed variations on the performance of real ball mills, and to evaluate the sensitivity of DEM predictions to various modelling assumptions.

An important observation is that changes in the charge composition lead to changes in the charge distribution inside the mill and the power draw via their effect on the shear strength of the charge. Changes to the PSD (both the lower cutoff of the size range and the shape of the probability distribution), the ball fraction and the rock shape distribution all alter the shear strength of the charge that is produced by the interlocking of these

particles. Increasing the angularity or aspect ratios of the rock or decreasing the proportion of smaller particles strengthens the charge and makes slope failure more difficult. For sub-critical speeds, this leads to higher dynamic angles of repose for the cascading free surface, to higher shoulder positions and to higher power consumption. For super-critical speeds, the stronger interlocking of particles reduces their mobility near the free surface of the centrifuged layer leading to significantly lower power consumption. Combined, these move the peak in the power curve to much lower and more realistic speeds.

As the particles become rounder (on average) or the proportion of smaller particles increases then the shear strength of the charge declines and slope failure occurs more easily leading to lower sub-critical power consumption. At super-critical speeds, the particles near the surface of the centrifuged layer flow more easily to positions lower in the mill leading to higher power consumption. This causes the peak power to move to higher speeds. This means that the location of the peak power is quite sensitive to the shear strength of the charge and therefore to the charge composition. This also means that predictions made using ball only DEM models differ substantially from those obtained when more realistic charge models are used.

The effect of the lifter wear cycle on the charge behaviour and power draw were studied. At sub-critical speeds, the wear induced increases in lifter face angle and reductions in lifter height lead to steadily decreasing amounts of cataracting material thrown shorter distances and higher toe positions. At low speeds, the steepest lifters produce the highest power draw. With increasing mill speed, the lower and shallower worn lifters produce higher power draw. For low mill speeds, the highest power draw is expected in the first half of the lifter life and then declines slowly. At higher speeds, the power draw initially increases as the lifters wear and then decreases. The face angle producing the highest power draw decreases steadily with increasing mill speed.

The effect of the fill level on mill behaviour was also analysed. For low mill speeds ($N \leq 70\%$) the highest specific power draw occurred for a fill level of only 10%. At a speed of 80% critical, the fill level producing the highest power input was 15% which then increases to 20% at $N = 90\%$ critical. The simulations also demonstrated that filling above 25–30% leads to significantly lower specific power consumption, since the additional material is closer to the axis of the mill and contributes little to the torque required to support the charge.

For mill speeds of $N \leq 70\%$, the power draw was found to be highly insensitive to the lifter pattern. For higher speeds, reasonable differences were observed with the hi–lo pattern having higher specific power draw than the standard hi–hi case and half the number of lifter having even higher specific power draws.

Changes to the ball fraction were found to have little effect on mill behaviour for rock fractions of 20% by volume or more. The power draw increases proportionally with the weight of the charge, but the specific power draw is insensitive to the ball fraction. For very small rock fractions, the specific power draw is consistently higher at sub-critical speeds. The ball density itself has almost no effect on the charge behaviour or on the specific power draw. For sub-critical speeds, particles below 5–10 mm seem to have little effect on the charge behaviour or the power draw. This suggests that truncating the PSD in DEM simulations of mills at these sizes is a reasonable modelling

assumption. For higher speeds, some sensitivity to the cutoff size is observed, but this is still small, being less than 2.5% for cutoffs of 5 mm or lower.

DEM modelling appears to be able to qualitatively capture the behaviour of the charge and to make sensible predictions of the power draw. The results given here demonstrate the effect of many of assumptions used in this type of modelling. Quantitatively accurate predictions require still better agreement with observations of such quantities as the speed at which the peak power occurs. Already we have demonstrated that the inclusion of particle shape plays an important role in improving quantitative agreement. Further work is required to validate these DEM models and to identify further improvements that will allow highly accurate predictions to be made. Future developments will include the direct prediction of particle size reduction, axial flows within the mill and the effect of the pulp.

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